

Overeducation Over the Lifecycle: Disentangling Frictions, Innate Ability, and Job-Specific Experience

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Abstract

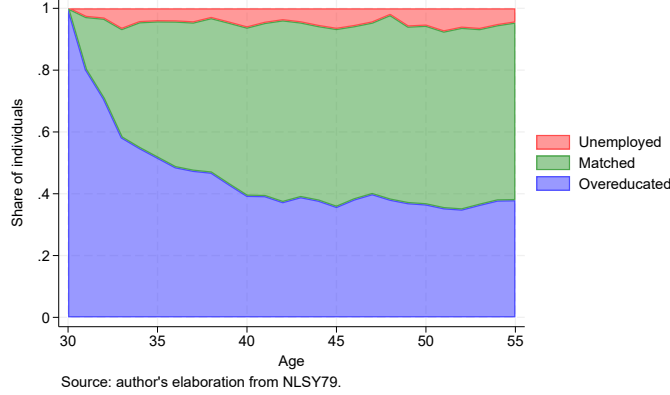
I study overeducation persistence with a directed-search model where workers differ in education, field, innate ability, job-specific experience, and age. Calibrated to the NLSY79 and O*NET, a structural decomposition shows that non-transferable job-specific experience is the dominant source of long-run persistence, frictions matter mostly early on, and slow ability learning amplifies both channels. Age effects and apparent overeducation are minor. Education is treated as exogenous to focus on post-schooling dynamics; selection is captured through heterogeneous ability distributions across education groups. Policies that speed early learning and reduce frictions are most effective.

1 Introduction

Why do some workers remain formally overqualified for their job for many years, even decades? Roughly 20–35% of workers globally hold an educational attainment higher than the typical requirement of the job they are employed in (Leuven and Oosterbeek (2011); McGuinness et al. (2018)), we refer to this condition as overeducation. However, most existing explanations treat overeducation as a transient mismatch that resolves as workers gain experience and find better-matched positions. This paper questions that premise. Figure 1 tracks the overeducation status of individuals who were overeducated at age 30 in the 1979 National Longitudinal Survey of Youth (NLSY79). While many initially transition into better matches — after one year around 20% exit overeducation, and after five years roughly half are no longer overeducated— approximately 40% remain overeducated after ten years, a share that barely changes over the following decade.

To investigate the mechanisms behind this persistence, I develop a directed search model, in the spirit of Menzio et al. (2016) and Menzio and Shi (2011), in which

Figure 1: Evolution of the status of overeducated workers at age 30.



Note: This figure follows all the individuals in the sample who are classified as *overeducated* at age 30 (NLSY79). The x-axis reports the age starting from 30; the y-axis shows the fraction of the original subsample (normalized to 100%) in each subsequent status. Series shown are: (i) **remaining overeducated** (blue area), that is, the share still employed in occupations whose typical educational requirement (from O*NET) is lower than the worker's attainment; (ii) **transitioned to matched** (green area), that is, the share employed in occupations whose typical requirement matches the worker's education; (iii) **transitioned to unemployment** (red area), that is, the share currently unemployed. Overeducation is defined as having attained a higher education level than the occupation's typical requirement. See Section 2.1 for sample restrictions and further details. See Section 2.3 for details on the overeducation definition.

workers differ in educational level and field of study, innate occupational ability, job-specific experience, and age. Directed search is the natural framework here: it allows workers to optimally target submarkets, trading off the generosity of a contract against the probability of finding a job. This makes the decision to remain in an overeducated position a rational, forward-looking choice rather than something that just happens to workers. On the firm side, occupations vary in their complexity, defined as the minimum skill level required for productive employment.

I calibrate the framework using US individual-level data from the NLSY79 and information on occupational requirements from O*NET¹. The model successfully replicates the empirical patterns of overeducation persistence observed in the data without directly targeting them in calibration. I then use the calibrated model to decompose persistence into four channels. First, labor market frictions, the “temporary” channel, account for roughly 25 percentage points of the total monthly persistence early in the career, falling to around 10 percentage points later in life: some overeducated workers are actively searching for a better match but have not yet found one. Second, and most importantly, the specialization channel accounts for 65–70 percentage points early in the lifecycle: workers accumulate job-specific, non-transferable experience in their

¹See Baley et al. (2022), Lise and Postel-Vinay (2020), and Guvenen et al. (2020) for similar strategies.

overeducated occupation, so that leaving would mean restarting with zero tenure in an unfamiliar role. This trap is amplified by ability learning frictions: the slower workers and firms learn about the worker’s suitability for an occupation, the greater the risk of switching, and hence the stronger the incentive to stay put. Third, the “age” channel, capturing skill depreciation and a shorter remaining horizon, plays only a minor role for younger workers, but together with accumulated experience becomes the primary driver of persistence in later career stages. Finally, a small share reflects “apparent overeducation” (Chevalier (2003)): workers who appear overeducated but are actually in their best feasible match. Here, this is driven by humanities graduates whose degree does not grant access to the most complex occupations.

This paper contributes mainly to three strands of the literature. First, it adds to the body of work on overeducation, dating back to Freeman (1975), who argued that returns to higher education declined in the 1970s. Much of the empirical work that followed, inspired by Duncan and Hoffman (1981), focused on the effects of overeducation on wages and employment (e.g., Verdugo and Verdugo (1989)). Given concerns about omitted variable bias (see Leuven and Oosterbeek (2011)), several studies attempted to address this using instrumental variables (e.g., Korpi and Tåhlin (2009)) or fixed effects (e.g., Bauer (2002); Dolton and Vignoles (2000)). However, the accuracy of these results is often questioned due to the difficulty of finding credible instruments and the possibility that workers sort into occupations based on unobserved characteristics (Leuven and Oosterbeek (2011)). Importantly, most of this literature has emphasized cross-sectional aspects (e.g., Alba-Ramirez (1993); Hartog (2000)), with relatively little attention to dynamics. Notable exceptions include Rubb (2003) and Clark et al. (2017), who provide empirical evidence of substantial persistence in overeducation. This observed persistence is hard to reconcile with existing theories. Human capital theory implies that all the investment in human capital sustained by individuals will eventually be rewarded by firms. The career mobility view (Sicherman and Galor (1990); Sicherman (1991)) suggests that workers accept initially mismatched jobs as stepping stones to better-matched positions, implying a temporary phenomenon. In a similar spirit, Dolado et al. (2009) develop a random search model in which workers, endowed with two skill levels and the option to search while employed (unlike in Gautier (2002)), may temporarily accept an overqualified position due to labor market frictions, but eventually transition to a matched job. None of these models, however, can explain the long-term persistence documented in this paper. This paper provides the first structural decomposition of overeducation persistence into its constituent channels over the lifecycle.

Second, this paper contributes to the literature on directed search theory, initiated by Montgomery (1991) and further developed by Moen (1997). The model builds closely

on Menzio et al. (2016), but introduces additional heterogeneity along three dimensions: workers differ not only in general experience, age, and employment status, but also in field of study, job-specific experience, and innate occupational ability. Each dimension does meaningful economic work. The field of study creates asymmetric access to high-complexity occupations and generates endogenous apparent overeducation. The fully non-transferable job-specific experience is the engine of the specialization trap. The ability learning process generates an option value of remaining in the current job that rises with tenure, reinforcing persistence.

Lastly, this paper is closely related to the literature on skills mismatch (e.g., Baley et al. (2022); Guvenen et al. (2020); Lise and Postel-Vinay (2020)). From this literature, I borrow the empirical measures of job complexity and workers' ability. Unlike these papers, which study the lifecycle dynamics of skill mismatch broadly, this paper focuses specifically on educational mismatch and uses a calibrated structural model to trace the channels through which it persists. Additionally, although the focus is on overeducation, the model can naturally accommodate the study of broader skill mismatch, since workers differ in their overall skill level as determined by education and innate ability.

The remainder of the paper is organized as follows. Section 2 describes the data and provides descriptive evidence. Section 3 presents the theoretical framework. Section 4 details the calibration, validates the model, and presents the decomposition results. Section 5 compares the baseline findings with those from the younger NLSY97 cohort, finding that overeducation persistence is even more pronounced for recent generations. Section 6 concludes.

2 Data and Descriptive Evidence

This section describes the data and documents the empirical patterns that motivate the model. Section 2.1 describes the primary dataset and sample construction. Section 2.2 explains how job complexity and workers' innate ability are measured. Section 2.3 defines education groups, fields of study, and the overeducation classification. Sections 2.6–2.11 present descriptive evidence on the incidence of overeducation and undereducation, the relationship between complexity and educational mismatch, the lifecycle patterns of transitions across employment states, and the age profile of wages by complexity and education.

2.1 Data and Sample

The primary dataset is the National Longitudinal Survey of Youth 1979 (NLSY79), a panel study that has followed a nationally representative sample of 12,686 U.S. citizens born between 1957 and 1964, interviewed annually from 1979 to 1994 and biannually thereafter. Its longitudinal structure allows me to track individuals' career paths over their working lives, which is essential for studying the long-term dynamics of overeducation.

I complement the NLSY79 with the O*NET database, which contains hundreds of standardized, occupation-specific descriptors for nearly 1,000 occupations covering the entire U.S. economy.

The analysis focuses on prime-age individuals between 25 and 55, covering the period when most workers have completed their formal education and are fully integrated into the labor market. Following standard sample construction (see Baley et al. (2022)), I drop individuals with more than two years of military service, weak labor attachment (more than ten years out of the labor market), those already working at the start of the sample period, and those with missing ability scores. The final sample consists of 3,297 individuals, yielding a total of 1,005,947 monthly observations.

Section 5 compares these baseline findings with those from the National Longitudinal Survey of Youth 1997 (NLSY97) to assess whether overeducation dynamics have remained consistent across generations.

2.2 Measuring Job Complexity and Workers' Innate Ability

To measure the complexity of each occupation, I follow the literature on skill mismatch (Baley et al. (2022), Lise and Postel-Vinay (2020), Guvenen et al. (2020)).

First, the complexity of each occupation is measured using detailed descriptors from the O*NET database. These descriptors provide comprehensive information on the skills and knowledge required for various jobs. Using a principal component analysis (PCA), the numerous raw descriptors are systematically reduced into four primary dimensions of complexity: verbal, mathematical, social, and technical skills. Then, each occupation is assigned to one of four complexity levels (r_1, r_2, r_3, r_4) based on the skill requirements in these four dimensions².

Second, workers' skills are measured using six Armed Services Vocational Aptitude Battery (ASVAB) scores from the NLSY79, together with individual scores on the Rotter locus of control scale and the Rosenberg self-esteem scale. Following the same dimensionality-reduction procedure, these scores are projected onto four dimensions of worker ability: mathematics, verbal, social, and technical skills. Each worker is

²See Appendix C for more details.

then assigned a binary innate ability status (high or low) in each occupation based on whether the distance between the worker’s skill vector and the occupation’s requirement vector falls below a specified threshold.³

2.3 Measuring Education, Fields, and Overeducation Status

Individuals are first divided into two educational groups: E^L , representing those with a high school degree or less, and E^H , for those with a college degree or higher (2 or more years of college completed). Among those with a college degree or higher, I further distinguish between two fields of study: humanities-related fields, E_{hum}^H , and non-humanities fields, E_{other}^H . I explain the rationale for this distinction below.

First, I assigned to each college graduate one of four possible fields: medical, humanities, social sciences and Law, STEM. Similarly, I assigned a field to each occupation based on the job family indicated in O*NET. As can be seen from Table 1, humanities is the only field that does not have any occupations in the most complex subset, r_4 . Therefore, to simplify the analysis, I group medical, social sciences and law, and STEM into a single field, assuming they have similar characteristics in terms of access to complex jobs.

Hence, the model distinguishes three worker types: low-educated (E^L), high-educated in humanities (E_{hum}^H), and high-educated in non-humanities fields (E_{other}^H). As will become clear in Section 3, E_{hum}^H workers are at a disadvantage relative to E_{other}^H because they cannot access the most complex occupations (r_4); at the same time, they hold an advantage over E^L workers in intermediate-complexity jobs (r_2, r_3), where their higher average skill level and the informativeness of their degree as an ability signal give them preferential access.

Finally, workers are classified as overeducated if their educational attainment is higher than the typical requirement for their occupation, as specified in the O*NET database. Similarly, workers are classified as undereducated (matched) if their education is lower than (equal to) the occupation’s typical requirement.

2.4 Incidence of Overeducation

As shown in Figure 2, approximately 16% of workers are overeducated at the beginning of their careers. This initial incidence gradually declines to about 10% by age 40 before slightly rising again later in life. In general, the incidence of overeducation is lower than found in other papers. This is likely to be a lower bound since only two educational levels (any college degree vs high school diploma or less) are used.

³See Section 3 for the theoretical definition of innate ability and Appendix C for full construction details.

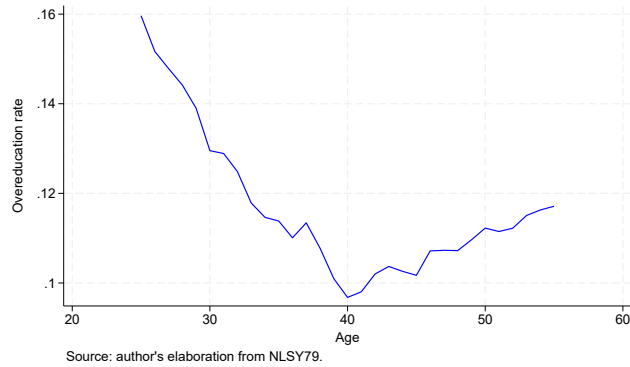
Table 1: Number of jobs by field and complexity level.

Job Field	r_1	r_2	r_3	r_4
Medical	1	3	10	11
Humanities	0	16	13	0
Social Science and Law	7	34	42	4
STEM	0	3	5	34
None	23	64	47	0
Total	31	120	117	49

Note: This table reports the number of occupations by job field and by complexity level (r_1, r_2, r_3, r_4). Complexity levels are based on Baley et al. (2022), who compute complexity scores in 4 dimensions (verbal, mathematical, social, technical) for each occupation using O*NET skill descriptors. Each cell contains the count of distinct occupations that belong to the indicated field and complexity level. The “Total” row reports column sums, that is, the total number of occupations with a given complexity. A zero entry means no occupations of that field are classified at that complexity. The “None” field groups occupations not assigned to one of the four main fields. See Section 2.2 for details on the complexity classification.

The decreasing pattern in the first half of the lifecycle is consistent with a model of frictions, learning, and sorting in which it takes time for workers to find the right job. The slight late-career increase is more puzzling: this paper argues it is driven primarily by workers accumulating non-transferable, job-specific experience in their current occupation – making a switch increasingly costly – rather than by age effects per se. I return to this in the formal decomposition of Section 4.3.

Figure 2: Share of overeducated workers by age.



Note: This figure reports the share of workers classified as overeducated at each age in the NLSY79 sample. Overeducation is defined by comparing individual education to the occupation’s typical requirement (O*NET). Series are unconditional cohort shares. See Section 2.1 for sample restrictions and further details. See Section 2.3 for details on the overeducation definition.

2.5 Overeducation and Complexity in the Data

To understand which types of jobs are most affected by overeducation, I examine the relationship between complexity and educational requirements. For occupations with low or intermediate complexity (r_1, r_2, r_3), a college degree is not a universal requirement, so these roles can in principle employ overeducated workers. Table 2 illustrates the gradient clearly: only about 6% of complexity-1 jobs and less than 17% of complexity-2 jobs formally require a college degree, whereas more than half of complexity-3 jobs do so.

Table 3 shows where overeducated workers are actually employed. The majority, nearly 70%, are concentrated in intermediate-complexity (r_2) occupations, with smaller shares in r_3 (around 22%) and r_1 (around 9%). This distribution indicates that overeducation is not a random phenomenon; it is disproportionately concentrated in occupations that pose a meaningful skill challenge but do not formally require a college degree.

Table 2: Education requirements and complexity levels.

Complexity	# occupations $E_j = E^H$	# occupations	Share $E_j = E^H$
r_1	2	31	6.45%
r_2	20	120	16.67%
r_3	63	117	53.85%
r_4	49	49	100%

Note: This table shows, for each complexity level r , the total number of occupations (# occupations) and the subset that formally requires a college degree ($E_j = E^H$). The percentages are computed as the ratio between occupations with a college requirement and the total occupations at the corresponding complexity level. Complexity levels are based on Baley et al. (2022), who compute complexity scores in 4 dimensions (verbal, mathematical, social, technical) for each occupation using O*NET skill descriptors. See Section 2.2 for details on the complexity classification.

Table 3: Overeducated workers and complexity.

Complexity (r)	# overeducated	Share out of total overeducated
1	10,533	8.88%
2	82,098	69.24%
3	25,937	21.88%
4	0	0%

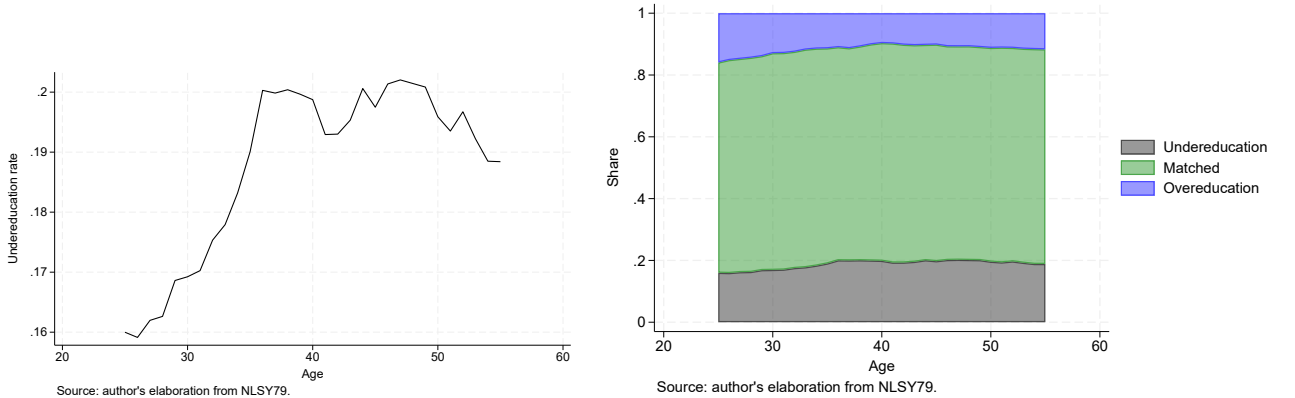
Note: This table reports the number and share of overeducated workers by occupation complexity (r_1, r_2, r_3, r_4), computed from the NLSY79 sample. Complexity levels are based on Baley et al. (2022), who compute complexity scores in 4 dimensions (verbal, mathematical, social, technical) for each occupation using O*NET skill descriptors. See Section 2.2 for details on the complexity classification. See Section 2.3 for details on the overeducation definition.

2.6 Incidence of Undereducation

The incidence of undereducation is substantial, ranging from 16% to over 20% across the lifecycle (Figure 3, left panel), a magnitude consistent with findings in the broader literature (Leuven and Oosterbeek (2011)). This confirms that education alone is not the decisive hiring criterion: other worker characteristics also matter.

More informatively, the share of undereducated workers rises steadily with age. This pattern is consistent with an ability-learning mechanism: as workers accumulate tenure, firms and workers gradually learn the worker's true innate ability in their occupation. When that ability is revealed to be high, lower-educated workers are more likely to stay in more complex jobs. At the same time, if ability is revealed to be low workers may move to other jobs (even if they have a college degree), producing a rising share of undereducation over the lifecycle. Figure 3 (right panel) provides the complete status distribution, showing overeducation, matched, and undereducation shares jointly for each age.

Figure 3: Share of undereducated workers by age (left panel). Share of workers in each status by age (right panel).



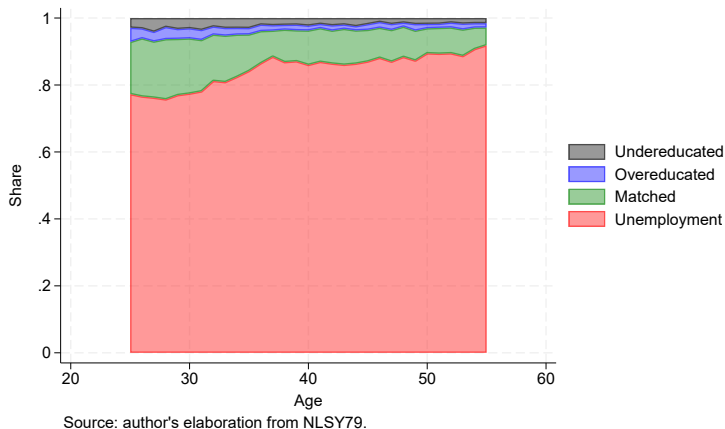
Note: This figure shows the share of undereducated workers by age (left panel) and the distribution of workers in each status: overeducation, matched or undereducation (right panel). Overeducation, undereducation and matched status are defined by comparing individual education to occupation requirements from O*NET. Series are unconditional cohort shares; see Section 2.1 for sample restrictions and measurement details. See Section 2.3 for details on the overeducation/undereducation/matched definitions.

2.7 Transitions from Unemployment

Figure 4 documents monthly transition rates out of unemployment. Younger unemployed workers exhibit substantially higher labor market fluidity: approximately 20% successfully find a job each month early in their careers, a rate that declines steadily with age. Among those who do find jobs, the majority accept a position that matches

their educational qualifications. Nevertheless, a non-trivial share still enters either an overeducated or undereducated role, reflecting the ongoing nature of labor market sorting even after a spell of unemployment. This suggests that short-term re-employment goals can take precedence over finding the ideal long-term match.

Figure 4: Monthly transition rates from unemployment by age.



Note: This figure presents monthly transition rates from unemployment to (i) matched (green area), (ii) overeducation (blue area), (iii) remaining unemployed (red area), and (iv) undereducation (black area). Rates are computed using the NLSY79 sample. Overeducation is defined as having attained a higher education level than the occupation's typical requirement. See Section 2.1 for sample restrictions and Section 2.3 for the overeducation/matched/undereducation definitions.

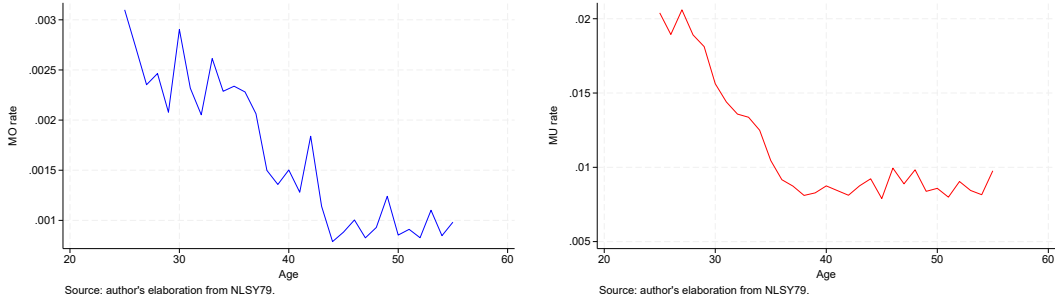
2.8 Transitions from Matched

Workers in matched jobs exhibit far more stable careers. As shown in Figure 5, the monthly transition rate from a matched job to an overeducated one is exceptionally low, fluctuating between 0.1% and 0.3% throughout the lifecycle. Once a worker secures a well-matched position, a regression into overeducation is very unlikely. Regarding transitions to unemployment, the monthly rate is around 2% early in the career and declines to less than 1% in the second half of the lifecycle.

2.9 Transitions from Overeducation

Figure 6 presents transition rates out of overeducation. Three patterns stand out. First, the monthly rate at which overeducated workers move to a matched job peaks at approximately age 32 and then declines rapidly, indicating that the early career is the critical window for sorting out of a mismatched position. Second, transitions to unemployment are most frequent among young workers (similarly to matched workers).

Figure 5: Monthly transition rates from matched to overeducated (left panel) and to unemployment (right panel).



Note: This figure shows monthly transition rates from matched employment to overeducation (left) and unemployment (right), by worker age. Rates are computed using the NLSY79 sample. Overeducation and matched statuses are defined by comparing individual education to occupation requirements from O*NET. See Section 2.1 for sample restrictions and Section 2.3 for the overeducation/matched definition.

Third, and most importantly, persistence — overeducated workers remaining overeducated — rise sharply with age (bottom panel). Together, these patterns point to two distinct mechanisms: an early-career sorting and learning process that corrects initial mismatches, and a late-career persistence trap in which the costs of leaving an overeducated role become prohibitive.

2.10 Overeducation Persistence

The central empirical fact of this paper is the monthly persistence rate of overeducation: the share of overeducated workers who remain in that status one month later. As shown in the bottom panel of Figure 6, this rate is far from constant. It stands at approximately 96.5% early in the career and rises to about 98.5% in later career stages.

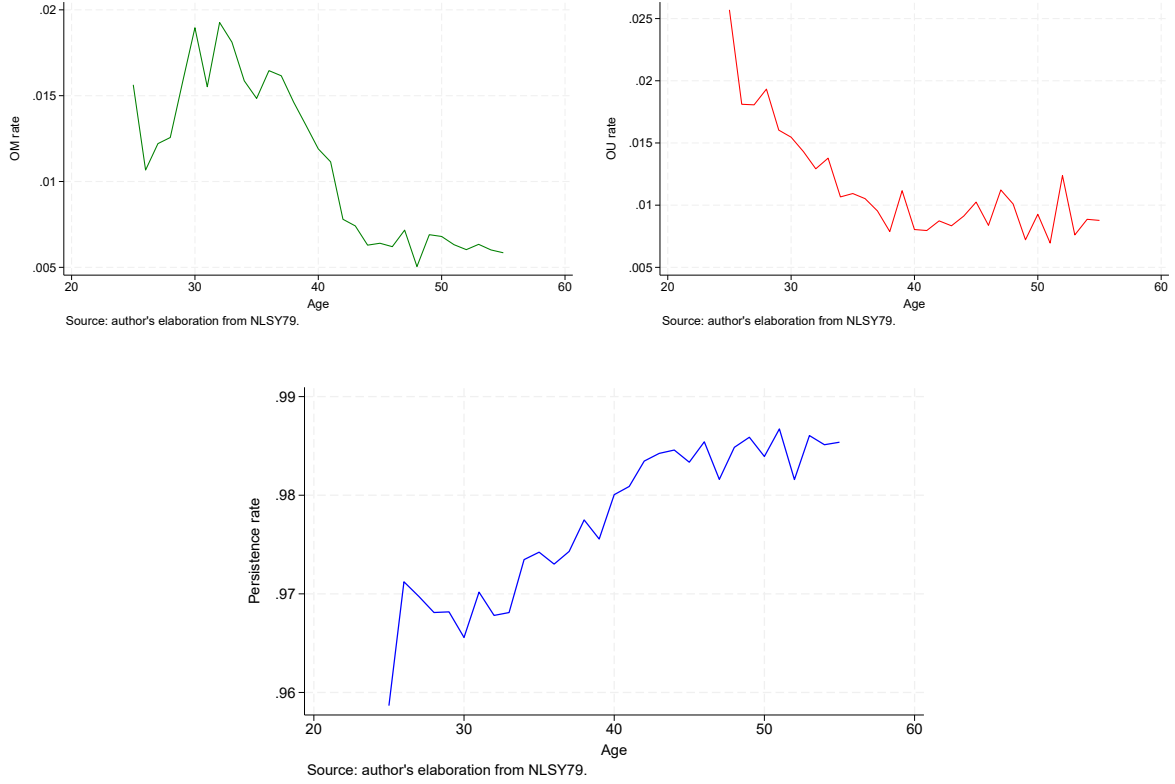
While overeducation may serve as a temporary sorting phase early in the career, it becomes increasingly permanent as workers age. The initial labor market frictions that generate mismatch are gradually resolved, but once a worker has accumulated job-specific experience in an overeducated role, the non-transferability of that experience and the rising costs of occupational switching keep persistence high. The structural decomposition in Section 4.3 quantifies the relative contribution of each channel.

2.11 Age Profile of Wages by Complexity and Education

This subsection documents the age profile of wages by job complexity and education group. The patterns provide descriptive evidence used to calibrate the model in Section 4.

Figure 7 reveals three salient facts. First, job complexity is the primary driver of

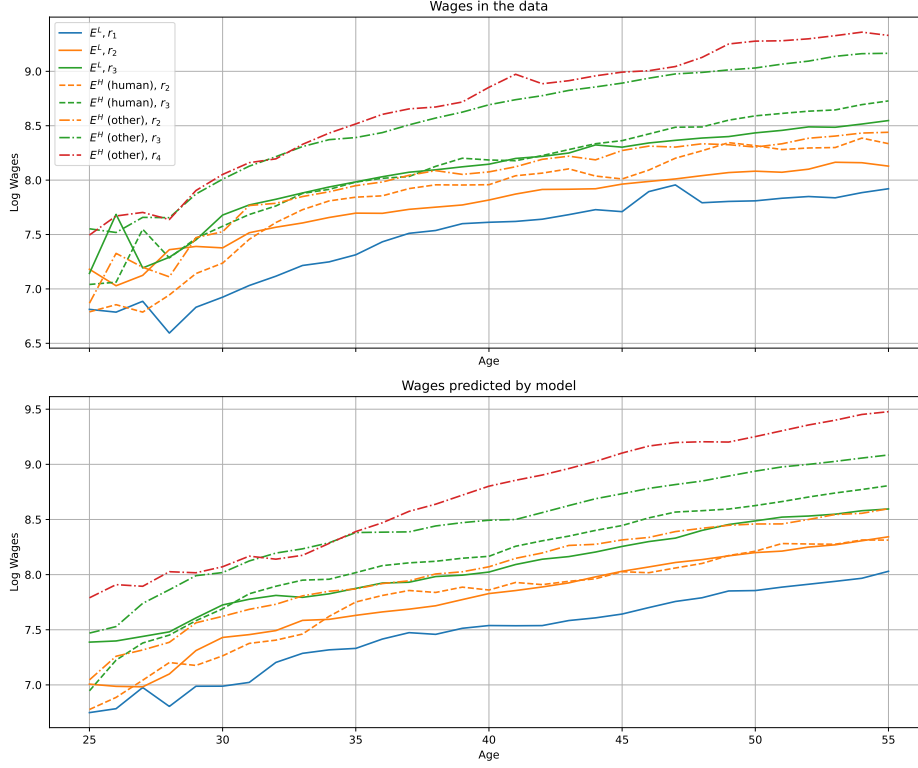
Figure 6: Monthly transition rates from overeducation to: matched (left panel), unemployment (right panel), and remaining overeducated (bottom panel).



Note: This figure presents monthly transition rates from overeducation to (i) matched (OM rate), (ii) unemployment (OU rate), and (iii) remaining overeducated (Persistence rate). Rates are computed using the NLSY79 sample. Overeducation is defined as having attained a higher education level than the occupation's typical requirement. See Section 2.1 for sample restrictions and Section 2.3 for the overeducation/matched definition.

wage levels: workers in r_4 occupations earn substantially more than those in r_1 at every age, with a near-monotone gradient across complexity levels. Second, wages follow a hump-shaped lifecycle profile for all groups, rising steeply early in the career before plateauing or slightly declining after mid-career. Third, conditional on complexity, education matters for wages: E_{other}^H workers earn more than E_{hum}^H workers in equivalent-complexity jobs, and the gap is largest in the highest-complexity occupations where the field-specific returns to a non-humanities degree are highest. This education-within-complexity wage gradient corroborates the field-specificity assumption introduced in Section 2.3 and motivates the separate treatment of humanities graduates in the model.

Figure 7: Wage profile of monthly log wages, by complexity and education. Actual data (top panel) vs wages implied by the calibrated model (bottom panel).



Note: This figure shows the age profile of monthly log wages by job complexity (different colour) and education level (different line style). E^L corresponds to workers with a high school diploma or less, while $E^H(human)$ corresponds to workers with a humanities college degree, and $E^H(other)$ corresponds to workers with other types of college degrees. The top panel displays actual data from the NLSY79 sample, while the bottom panel presents wages implied by the calibrated model. Complexity levels are based on Baley et al. (2022), who compute complexity scores in 4 dimensions (verbal, mathematical, social, technical) for each occupation using O*NET skill descriptors. See Section 2.2 for details on the complexity classification. See Section 4 for details on the model calibration.

3 Theoretical Framework

This section develops the directed search model used to study overeducation persistence. Section 3.2 defines worker skill as a function of education and innate ability. Section 3.3 introduces field-specificity of education. Section 3.4 maps skill levels to occupation complexity. Section 3.5 describes the stochastic education assignment. Section 3.7 derives the equilibrium educational requirement set by firms. The full dynamic model is presented in Sections 3.6–3.9.

3.1 The Environment

The economy consists of T overlapping generations. Time is discrete, and $t \in \{0, 1, \dots, T\}$ denotes the age of the worker. Workers and firms are risk-neutral. Each worker is en-

dowed with one indivisible unit of labor and maximizes the expected discounted sum of per-period consumption with discount factor $\beta \in (0, 1)$. A continuum of firms, with positive measure, populate the economy.

There are J occupations, indexed by $j \in \{1, 2, \dots, J\}$, differing in their *complexity* r_j , defined as the minimum skill level required for productive employment. Worker i 's experience in occupation j is $e_{ij} \in \{0, 1, \dots, t\}$, the number of past periods spent employed in that occupation. Experience is fully occupation-specific: switching to a new occupation resets experience to zero. This non-transferability is the key mechanism behind the specialization trap: as overeducated workers accumulate tenure, the cost of switching occupations rises, which is the primary source of long-run persistence in the model. A robustness check in Appendix A allows 50% experience portability; the main results are unchanged.

The output of a firm–worker match is

$$y(r_j, e_{ij}, s_{ij}, t) = \begin{cases} g(r_j, e_{ij}, t) & \text{if } s_{ij} \geq r_j \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where $g(\cdot)$ is increasing in complexity r_j and experience e_{ij} , and decreasing in age t (to capture skill depreciation: productivity loss due to aging). Conditional on satisfying the skill requirement and given (r_j, e_{ij}, t) , all workers are equally productive: a worker with $s_{ij} > r_j$ possesses skills that are simply not utilized in that job, and those surplus skills do not raise output.⁴

3.2 Skill Level of the Worker

The skill level of a worker i in occupation j is given by

$$s_{ij} = s(a_{ij}, E_i)$$

where $E_i \in \{E^L, E_f^H\}$ denotes the worker's education. Education is publicly observable and can be either low (E^L), corresponding to a high school diploma or less, or high (E_f^H), corresponding to a college degree in field f . Importantly, a college-educated worker in field f can benefit from higher education only in occupations that belong to the same field. In occupations associated with a different field ($f_i \neq f_j$), there is no distinction between E_f^H and E^L . The role of fields is further discussed in Section 3.3.

The second component of skill is innate ability in occupation j , denoted $a_{ij} \in \{a^L, a^H\}$. This can be interpreted as the match-specific component of productivity

⁴This is a tractability assumption. There is evidence that more educated workers are not systematically more productive at a given complexity level (Gautier et al. (2002)).

between worker i and firms in occupation j . Ability is initially unobserved by both parties (see Baley et al. (2022) for evidence that workers themselves lack full information about their own occupational aptitude). Each period, with probability α , the true value of a_{ij} is publicly revealed; until then it is denoted $a_{ij} = a^0$. This learning friction is a key amplifier of persistence: while ability remains unknown, switching occupations carries the risk of discovering low ability in the new job, which raises the option value of staying put.

With binary education and binary ability, four skill levels arise, ordered as follows:

$$s(a^L, E^L) \leq s(a^L, E_f^H) \leq s(a^H, E^L) \leq s(a^H, E_f^H).$$

The ordering of the intermediate levels — placing low-ability college graduates below high-ability high school graduates — is supported by two facts. First, lifetime earnings show high-ability non-college workers out-earn low-ability college graduates (Ichino et al. (2024)). Second, the reverse ordering would imply a clean separation between educational levels, leaving no room for the widespread undereducation documented in Section 2.6 and the literature (Leuven and Oosterbeek (2011)).

3.3 Fields of Study

Each occupation j is associated with a field f_j . A college-educated worker with degree field f (E_f^H) benefits from higher education only in occupations in the same field ($f_i = f_j$). In cross-field matches ($f_i \neq f_j$), firms perceive no difference between E_f^H and E^L .

For example, a philosophy graduate applying to an engineering role is treated identically to an applicant without a college degree; conversely, an engineering graduate gains no advantage in philosophy-related jobs. The value of a degree therefore depends on degree level *and* field–occupation alignment.

In the empirical part of the paper, as discussed in Section 2.3, I distinguish between humanities graduates, who cannot access the most complex occupations, generating a structural disadvantage relative to non-humanities graduates and producing a form of apparent overeducation (Chevalier (2003)).

3.4 Occupations and Job Complexity

Each occupation requires a minimum skill level r_j , which I refer to as the complexity of the job. Since there are four distinct skill levels, determined by education and innate ability, there are also four corresponding complexity levels: $r_1 \leq r_2 \leq r_3 \leq r_4$. The following matrix illustrates the four possible skill levels of the worker in a given occupation, along with the complexity levels where the minimum skill requirement is

satisfied.

		Innate Ability	
		a^L	a^H
Education	E^L	$s(a^L, E^L)$ Req. satisfied: r_1	$s(a^H, E^L)$ Req. satisfied: r_1, r_2, r_3
	E^H	$s(a^L, E_f^H)$ Req. satisfied: r_1, r_2	$s(a^H, E_f^H)$ Req. satisfied: r_1, r_2, r_3, r_4

Note: This table summarizes the mapping between worker education (E^L, E_f^H), innate ability (a^L, a^H), and occupation complexity levels (r_1, r_2, r_3, r_4). It shows which education-ability combinations satisfy each complexity requirement and is intended to be a self-contained guide to the skill taxonomy used in the model. For example, a worker with low education and high ability (top-right entry) meets the skill requirements for occupations with complexity levels r_1, r_2 , and r_3 , whereas she will not be productive in jobs with complexity r_4 (the minimum requirement is high ability and high education). Remember that a given worker has a different ability level a_{ij} for each occupation j .

This implies that the set of occupations can be partitioned into four distinct complexity groups, each associated with a minimum skill level required for successful job performance.

Occupations with complexity r_1 are the least demanding and can be performed by all workers, since even the lowest education-ability combination suffices. By contrast, occupations with complexity r_4 are the most demanding and require both high education in the same field of the occupation and high innate ability. Occupations characterized by complexity r_2 require high education or high ability: for instance, a worker with $s(a^L, E_f^H)$ or $s(a^H, E^L)$ would meet the requirement. Finally, in jobs with complexity r_3 high ability is strictly necessary. Education per se is not required to meet the skill requirement, but it may serve as a signal, as discussed in Section 3.7.

Each group of occupations can include jobs tied to different educational fields. Hence, the match between a worker and an occupation depends not only on education and ability but also on the field-specific alignment.

3.5 Educational Choice

Since the focus is post-schooling dynamics, I treat education as stochastically assigned before entry. The probability of completing higher education in field f increases in the individual's average innate ability across occupations, $\bar{a}_i = \frac{1}{J} \sum_{j=1}^J a_{ij}$, capturing sorting without an explicit choice model.

Conditional on field–occupation alignment, college graduates are more likely to have high innate ability:

$$P(a_{ij} = a^H \mid E_{f_j}^H) \equiv \bar{a}(E_{f_j}^H) > \bar{a}(E^L) \equiv P(a_{ij} = a^H \mid E^L) = P(a_{ij} = a^H \mid E_{f' \neq f_j}^H).$$

Education thus serves two roles: it raises skill via human capital accumulation (see Section 3.2) and signals higher expected innate ability, but only in field-matched occupations.

This framework generates persistent overeducation: search frictions prevent immediate matching, and once stuck, specialization and learning channels amplify persistence.

3.6 The Labor Market

The type of a worker is summarized by $(E_i, e_{ij}, r_j, a_{ij}, t) = (\boldsymbol{\omega}, t)$, that is, job-specific experience e_{ij} , complexity r_j and ability $a_{ij} \in \{a^0, a^L, a^H\}$ in the current (if employed) or most recent (if unemployed) occupation, education E_i and age t . Each vacancy must specify $(\boldsymbol{\omega}, x, t)$, where x is the value offered to the worker in lifetime utility. Search is directed, meaning that there is a continuum of submarkets indexed by $(\boldsymbol{\omega}, x, t)$. Workers decide optimally where to search based on their type, and firms decide how many vacancies to create and where to locate them. $\theta_t(\boldsymbol{\omega}, x)$, the ratio between the vacancies and the individuals searching in a certain submarket, is the market tightness of the submarket $(\boldsymbol{\omega}, x, t)$. Workers and vacancies in the same submarket meet through a frictional process. Searching on-the-job is also possible: while employed workers are allowed to search for a job in a different occupation, but at a lower intensity than unemployed workers.

3.7 Educational Requirement

This section describes how firms determine the minimum educational requirement for each vacancy when a worker’s innate ability is unknown (i.e., $a_{ij} = a^0$). Since workers who satisfy the skill requirement $s_{ij} \geq r_j$ are equally productive, firms are only interested in maximizing the probability that a matched worker satisfies the requirement. Therefore, for a given occupation j , firms base their educational requirement E_j on the conditional probability $P(s_{ij} \geq r_j \mid E_i)$: the probability that a worker’s skill level s_{ij} , which depends on education E_i and innate ability a_{ij} , is greater or equal than the minimum skill requirement r_j .

For jobs with the lowest complexity r_1

$$P(s_{ij} \geq r_1 \mid E^L) = P(s_{ij} \geq r_1 \mid E^H) = 1$$

Thus, all workers satisfy the skill requirement regardless of their education and innate ability. So, the minimum educational requirement is $E_j = E^L$.

On the other hand, for the most complex occupations r_4

$$P(s_{ij} \geq r_4 | E^L) = 0 < P(s_{ij} \geq r_4 | E_{f_j}^H) = P(a_{ij} = a^H | E_{f_j}^H) \equiv \bar{a}(E_{f_j}^H)$$

Therefore, the minimum educational requirement in this case is $E_j = E_{f_j}^H$.

For jobs with complexity r_3 , firms are not interested in education per se, but it can be a signal for high ability, more specifically

$$P(s_{ij} \geq r_3 | E^L) = P(a_{ij} = a^H | E^L) \equiv \bar{a}(E^L) < \bar{a}(E_{f_j}^H) \equiv P(a_{ij} = a^H | E_{f_j}^H) = P(s_{ij} \geq r_3 | E_{f_j}^H)$$

So in this case, firms may decide to open a vacancy for the same occupation, either directed at low-educated or high-educated workers, at potentially different conditions.

Similarly, for jobs with complexity r_2

$$P(s_{ij} \geq r_2 | E^L) = P(a_{ij} = a^H | E^L) \equiv \bar{a}(E^L) < 1 = P(s_{ij} \geq r_2 | E_{f_j}^H)$$

Also in this case, a vacancy may specify different educational requirements.

3.8 Timing, State and Contracts

The model follows Menzio et al. (2016), adapted to accommodate the heterogeneity described above.

Aggregate state. At the beginning of each period, the aggregate state is $\psi = (n, u, e, \gamma)$, where $n(E, t)$ is the measure of non-participants, $u(\omega, t)$ the measure of unemployed, $e(\omega, t)$ the measure of employed, and γ the current realization of the cohort-size process.

Within-period timing. Each period has five stages: entry-and-exit, separation, search, matching, and production.

Entry and exit. A non-participating worker of age t enters the labor market with education and ability drawn as in Section 3.5. Participating workers may exit permanently.

Separation. An employed worker becomes unemployed with probability $d \in [\delta, 1]$, where $\delta \in [0, 1]$ is the exogenous destruction rate and d is set by the employment contract.

Search. Unemployed workers search with probability $\lambda_u \in (0, 1]$; employed workers

who retained their job search on-the-job with probability $\lambda_e \in (0, 1]$; workers who just lost their job cannot search in the current period. When searching, a worker optimally selects a submarket. Firms choose how many vacancies to post in each submarket at a flow cost $k > 0$ per vacancy.

Matching. In submarket (ω, x, t) , a searching worker meets a vacancy with probability $p(\theta_t(\omega, x))$, where $p : \mathbb{R}_+ \rightarrow [0, 1]$ is twice-differentiable, strictly increasing, strictly concave, with $p(0) = 0$ and $p(\infty) = 1$. A vacancy meets a worker with probability $q(\theta) = p(\theta)/\theta$, $q(0) = 1$, $q(\infty) = 0$. Upon meeting, the firm offers a contract worth x in lifetime utility; the worker accepts or returns to her previous status.

Production. An unemployed worker produces and consumes b . An employed worker produces $y(r, e, s, t)$ and receives wage w as specified in her contract. Ability is revealed with probability $\alpha \in (0, 1]$. At the end of the period, nature draws next period's cohort size from $\Pi(\hat{\gamma}|\gamma)$. Hats denote next-period variables throughout.

Contracts. Employment contracts are complete: they specify the wage w , the endogenous separation probability d , and the submarket where the employee should search, all as functions of the match history and the aggregate state. Firms therefore offer bilaterally efficient contracts that maximize the joint surplus of the match (see Menzio and Shi (2011)).

3.9 Characterization of the Equilibrium

The value of unemployment for a worker of type $(\omega, t) = (E_i, e_{ij}, r_j, a_{ij}, t)$ is

$$U_t(\omega, \psi) = b + \beta \mathbb{E}_{\hat{\psi}|\psi} [U_{t+1}(\omega, \hat{\psi}) + \lambda_u R_{t+1}(\omega, \hat{\psi})]$$

$$R_{t+1}(\omega, \hat{\psi}) = \max \left\{ \max_x \{p(\theta_{t+1}(\omega, x, \hat{\psi}))[x - U_{t+1}(\omega, \hat{\psi})]\}, \max_{x,r} \{p(\theta_{t+1}(\omega_0, x, \hat{\psi}))[x - U_{t+1}(\omega, \hat{\psi})]\} \right\}$$

where $(\omega_0, t) = (E_i, e_{ij} = 0, r, a^0, t)$ is the type of the worker who accepts a job of complexity r she has never performed in the past. In the current period, the unemployed produces and enjoys b (home production, leisure, unemployment benefit). In the following period, with probability λ_u the worker can search into the labor market, where she can choose whether to search for a job in the most recent occupation (first term of R_{t+1}) or a new career with no experience and unknown ability (second term of R_{t+1}); in both cases the optimal submarket is chosen by maximizing the probability of finding a job, $p(\theta)$ (decreasing in x), times the surplus deriving from the new employment condition, $x - U_{t+1}$.

The sum of the firm's lifetime profits and the worker's lifetime utility in a match

with known $a \neq a^0$ is

$$\begin{aligned} V_t(\boldsymbol{\omega}(a), \psi) = & y(r_j, e_{ij}, s_{ij}, t) + \beta \mathbb{E}_{\hat{\psi}|\psi} \left[\max_{d \in [\delta, 1]} \{ dU_{t+1}(\hat{\boldsymbol{\omega}}(a), \hat{\psi}) + \right. \\ & \left. + (1-d)[V_{t+1}(\hat{\boldsymbol{\omega}}(a), \hat{\psi}) + \lambda_e S_{t+1}(\hat{\boldsymbol{\omega}}(a), \hat{\psi})] \} \right] \\ S_{t+1}(\hat{\boldsymbol{\omega}}(a), \hat{\psi}) = & \max_{x, r} \{ p(\theta_{t+1}(\boldsymbol{\omega}_0, x, \hat{\psi})) [x - V_{t+1}(\hat{\boldsymbol{\omega}}(a), \hat{\psi})] \} \end{aligned}$$

where $\hat{\boldsymbol{\omega}}(a) = (E_i, \hat{e}_{ij}, r_j, a, t)$ and $\hat{e}_{ij} = e_{ij} + 1$. In the current period, the employed produces y , which depends on whether the skill requirement is satisfied, $s_{ij} \geq r_j$, the experience of the worker in the current occupation, e_{ij} , the job complexity, r_j , and the age of the worker, t . In the following period, with probability d the match is destroyed and the worker moves to unemployment. Otherwise, the match continues, but the possibility of search on-the-job must be taken into account. In particular, an employed worker has the opportunity to search for a different job with probability λ_e . In this case, the worker optimally chooses the submarket where to search, with no experience and unknown ability.

The sum of the firm's lifetime profits and the worker's lifetime utility in a match with unknown ability $a = a^0$ is

$$\begin{aligned} V_t(\boldsymbol{\omega}(a^0), \psi) = & \alpha \mathbb{E}[V_t(\boldsymbol{\omega}(a), \psi) | E_i] + (1-\alpha) \mathbb{E}[y(r_j, e_{ij}, s_{ij}, t) | E_i] + \\ & + (1-\alpha) \beta \mathbb{E}_{\hat{\psi}|\psi} \max_{d \in [\delta, 1]} \{ dU_{t+1}(\hat{\boldsymbol{\omega}}(a^0), \hat{\psi}) + \\ & + (1-d)(V_{t+1}(\hat{\boldsymbol{\omega}}(a^0), \hat{\psi}) + \lambda_e S_{t+1}(\hat{\boldsymbol{\omega}}(a^0), \hat{\psi})) \} \end{aligned}$$

In the current period, with probability α agents discover the true value of the innate ability of the worker in the current occupation, a_{ij} , otherwise they have an expectation of the output based on education. The following period is equivalent to the previous value function. Let's write explicitly the two expectation terms

$$\begin{aligned} \mathbb{E}[V_t(\boldsymbol{\omega}(a), \psi) | E_i] = & \bar{a}(E_i) V_t(\boldsymbol{\omega}(a^H), \psi) + (1 - \bar{a}(E_i)) V_t(\boldsymbol{\omega}(a^L), \psi) \\ \mathbb{E}[y(r, e, s, t) | E_i] = & \bar{a}(E_i) y(r, e, s(a^H), t) + (1 - \bar{a}(E_i)) y(r, e, s(a^L), t) \end{aligned}$$

where $\bar{a}_E(E_i) = P(a_{ij} = a^H | E_i)$ is the probability that a worker has high ability in a certain occupation, which depends on the educational level.

Market tightness should be such that

$$k \geq q(\theta_t(\boldsymbol{\omega}, x, \psi)) [V_t(\boldsymbol{\omega}, \psi) - x]$$

and $\theta_t(\boldsymbol{\omega}, x, \psi) \geq 0$ with complementary slackness. This condition ensures that firms

behave optimally since, in submarkets with at least a vacancy, the cost of maintaining open a vacancy, k , must equal the expected benefit of an open vacancy (RHS), that is the probability of meeting a worker, $q(\theta)$, times the surplus extracted by the firm, $V - x$. If instead a submarket is empty ($\theta = 0$), it must be that the cost of the vacancy is greater than the expected benefit for the firm.

The unique equilibrium is block-recursive (BRE): a recursive equilibrium in which the agents' value and policy functions do not depend on the aggregate state of the economy ψ . The proof is reported in Appendix B and follows Menzio et al. (2016) closely.

4 Calibration, Validation and Decomposition of Overeducation Persistence

This section brings the model of Section 3 to the data. Section 4.1 describes the calibration strategy and reports the parameter estimates. Section 4.2 validates the model by showing that it replicates out-of-sample the lifecycle pattern of overeducation persistence. Sections 4.3 and 4.4 present the structural decomposition. Sections 4.5 and 4.6 examine the mechanism of learning frictions in detail, and Section 4.7 characterizes the optimal behavior of college graduates in intermediate-complexity jobs.

4.1 Calibration

Production function and functional forms. I specify the production function as

$$g(r_j, e_{ij}, t) = \rho_r ((1 - \phi_1) + \phi_1 e_{ij}^{\phi_2}) - \phi_3 t,$$

where ρ_r scales output by job complexity (e.g., ρ_1 for r_1), the parenthetical term is an increasing, concave function of occupation-specific experience, and $\phi_3 t$ captures skill depreciation. Following Menzio et al. (2016), the matching probability takes the form $p(\theta) = \min\{\theta^{1/2}, 1\}$.

The model has sixteen free parameters: the discount factor β ; the high-ability probabilities $\bar{a}(E^L)$, $\bar{a}(E_{hum}^H)$, $\bar{a}(E_{other}^H)$, which govern education's signaling role; the search intensities λ_u and λ_e ; the unemployment flow value b ; the exogenous separation rate δ ; the ability learning probability α ; the vacancy flow-cost k ; and the seven production-function parameters $\rho_1, \rho_2, \rho_3, \rho_4, \phi_1, \phi_2, \phi_3$.

Step 1: Production function (wages). The seven production-function parameters are calibrated by minimizing the distance between model-implied average log wages

and their empirical counterparts from Section 2.11. I assume wages are a constant fraction of match productivity, allowing the fraction to differ across education groups.⁵ Figure 7 confirms that the calibrated model replicates the lifecycle wage profiles across all complexity–education cells with reasonable accuracy.

Step 2: Assigned and normalized parameters. Several parameters are assigned directly from external sources or normalization. λ_u is normalized to 1. The discount factor β is set so that $\beta^{-12} - 1 = 4.04\%$, the average Federal Funds Rate between 1982 and 2020. The exogenous separation rate $\delta = 0.012$ is taken from Baley et al. (2022), who estimate it using the same NLSY79 data. The unemployment flow value b is calibrated from the wage-replacement ratio $w/b = 0.71$ estimated by Hall and Milgrom (2008). The high-ability probabilities $\bar{a}(E_i)$ are assigned using the NLSY79 computed ability scores (see Section 2.2): for each education group E_i , I use the average share of occupations in which a worker’s ability is high a^H .

Step 3: Search and learning parameters (targeted moments). The three remaining parameters, α, λ_e, k , are those related to search and are calibrated by targeting some moments, as I describe in the following lines. First of all, I focus on workers younger than 35 years old. The reason is that young workers are more mobile across occupations and have a more pronounced learning process, making them more informative for the calibration of these parameters. Including older workers would make the calibration of the search parameters more difficult.

The monthly learning probability, α , is calibrated by targeting the ratio between the separation rate after the third and the eighteenth month of job-specific experience, $\log(haz_3/haz_{18})$, which Baley et al. (2022) estimated to be 1.37. To isolate the learning process, I restrict (only for the calibration of α) the attention to low-educated workers who are employed in jobs with complexity r_2 . The reason why I chose this group of people is that, in the model, when low-educated workers are employed in a job with complexity r_2 , the ability learning process is crucial in determining whether their match will continue or not⁶. Therefore, the separation rate of these kinds of matches after different months of experience is very informative about the speed of the learning process of the model. To calibrate the search effort during employment, λ_e , I target the average transition from one occupation to another (EE rate). Intuitively, since in the model on-the-job search is possible only across different occupations, the rate at which workers change occupation is informative about the search effort while employed.

⁵This assumption is consistent with the wage-posting structure of the directed search equilibrium.

⁶If the ability of the worker turns out to be low, the match is revealed to be not productive and ends immediately.

Lastly, I use the average transition rate from unemployment to employment (UE rate) to calibrate the vacancy cost, k .

Table 4 displays the target and calibrated moments, which are quite close for the UE rate and the separation rate ratio. As far as transitions from one occupation to another are concerned, the model predicts a lower EE rate; this is likely due to the fact that the empirical EE rate embeds mechanisms that are outside of the model. For example, technological change and automation, or people may move to a similar occupation without changing their overeducation status. However, these factors are not related to the persistence of overeducation and they do not invalidate the calibration strategy and the results of the paper. Table 5 shows all the calibrated parameters of the model. The calibrated ability learning parameter (α) implies the monthly probability that agents discover the true ability of the worker in the current occupation is around 14%. This means that after 3 months, ability is unknown in 63.6% of the matches; after 6 months, this share drops to approximately 40%, and after a year, 16% of the agents are involved in matches with unknown ability. Therefore, learning frictions play a non-negligible role in shaping the optimal decisions of the workers. The calibrated high-ability probability suggests there is a high degree of selection into education. Indeed, low-educated workers have 41.1% probability of being $a_{ij} = a^H$ in a certain occupation, whereas for humanities graduates this probability increases to 66.9% probability, followed by an even higher probability for non-humanities graduates, 73.4%. Finally, the production function calibrated parameters imply that jobs with complexity r_3 and r_4 are substantially more productive than less complex jobs, suggesting that the high-ability requirement is linked to more productive jobs. At the same time, the productivity in the most complex jobs (r_4) is only slightly higher than jobs with r_3 , indicating that there is not a strong incentive to move to a more complex occupation when employed in a r_3 complexity job.

Table 4: Model vs. Data moments.

Moment	Model	Data	Source
UE rate	0.2087	0.2099	NLSY79
EE rate	0.0048	0.0188	NLSY79
$\log(\text{haz}_3)/\log(\text{haz}_{18})$	1.3706	1.37	Baley et al. (2022)

Note: This table reports the calibration targets and the corresponding model-generated moments. “UE rate” denotes the monthly unemployment-to-employment transition rate; “EE rate” denotes the monthly employment-to-employment occupation-change rate; and $\log(\text{haz}_3)/\log(\text{haz}_{18})$ is the log ratio of separation hazards at 3 and 18 months of tenure. Data sources and sample restrictions are detailed in Section 2.1. Calibrated moments are generated using the theoretical transition probabilities of the model. For example, the UE rate is given by $\lambda_u p(\theta(\omega, t))$, whereas the EE rate is given by $(1 - d(\theta(\omega, t)))\lambda_e \max_{x,r} p(\theta(\omega_0, x, t))$.

Table 5: Calibrated parameters.

Parameter	Description	Value	Methodology
b	Unemployment flow value	5.3	Match $b/w = 0.71$, Hall and Milgrom (2008)
λ_u	Search intensity when unemployed	1	Normalized
λ_e	Search intensity while employed	0.8390	Match EE rate, calibrated
δ	Exogenous separation rate	0.012	Assigned, Baley et al. (2022)
α	Ability learning probability	0.1395	Match separation rate ratio, calibrated
k	Vacancy flow-cost	299.28	Match UE rate, calibrated
β	Discount factor	0.9967	Match average interest rate, assigned
ρ_1	Production function: complexity 1	10.938	Calibrated with age-profile of wages
ρ_2	Production function: complexity 2	11.377	Calibrated with age-profile of wages
ρ_3	Production function: complexity 3	11.88094	Calibrated with age-profile of wages
ρ_4	Production function: complexity 4	11.88104	Calibrated with age-profile of wages
ϕ_1	Production function	3.911	Calibrated with age-profile of wages
ϕ_2	Production function	0.03435	Calibrated with age-profile of wages
ϕ_3	Skill depreciation	0.00022	Calibrated with age-profile of wages
$\bar{a}(E^L)$	Probability of high ability: E^L	0.4112	NLSY79, assigned
$\bar{a}(E_{hum}^H)$	Probability of high ability: E_{hum}^H	0.669	NLSY79, assigned
$\bar{a}(E_{other}^H)$	Probability of high ability: E_{other}^H	0.734	NLSY79, assigned

Note: This table reports the calibrated parameters, their brief description, calibrated values, and the identification method or data source used. See Section 4 for details on the calibration procedure and targeted moments.

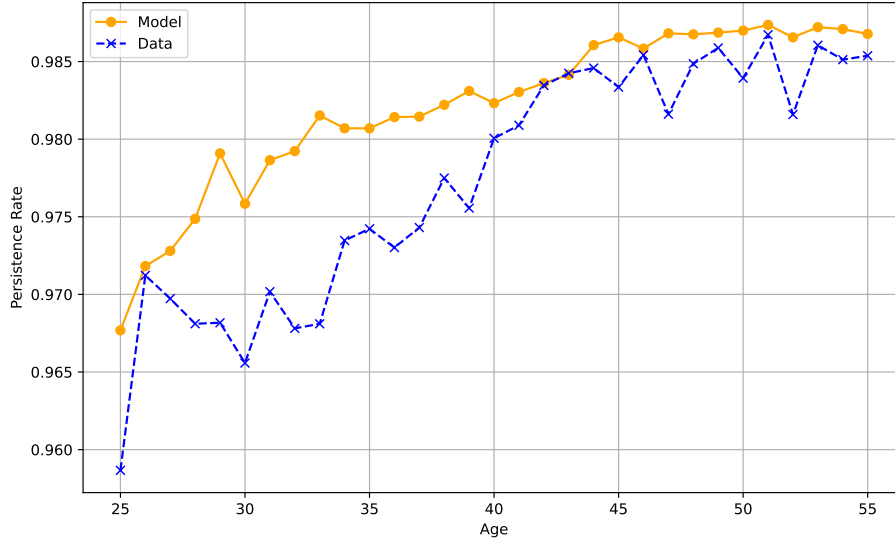
4.2 Validation: Overeducation Persistence in the Model

As a form of validation, Figure 8 provides a critical comparison, showing the monthly overeducation persistence rates from both the calibrated model and the NLSY79 data. The model successfully replicates the empirical pattern, capturing the initial, lower persistence rate early in the lifecycle and its subsequent increase with age. While the empirical fit of wages (Figure 7) was directly targeted, nothing directly related to the overeducation persistence has been used in the calibration strategy. So, the fact that the magnitude and the pattern of the persistence in the model is closely aligned with the data is not obvious and should be seen as an important result for the theory.

4.3 Decomposition of Overeducation Persistence: Methodology

This and the following section are dedicated to the key objective of this paper: decomposing the monthly persistence of overeducation into the main channels of the model. In the following lines, I will briefly recap the meaning of each channel and explain how I assign a certain channel to each overeducated worker in the dataset. First, the “temporary” channel is assigned when workers are searching for an alternative occupation, but they did not manage to find a new match due to search frictions. This is

Figure 8: Overeducation monthly persistence: Model vs Data.



Note: This figure compares monthly overeducation persistence rates from the calibrated model and from the NLSY79 data. The dashed line represents the empirical persistence (the same as in Figure 6). The solid line is the persistence predicted by the model. Overeducation is defined as having attained a higher education level than the occupation’s typical requirement (O*NET). Persistence is the share of overeducated workers who remain overeducated one month later. The persistence rate is not directly targeted in the calibration.

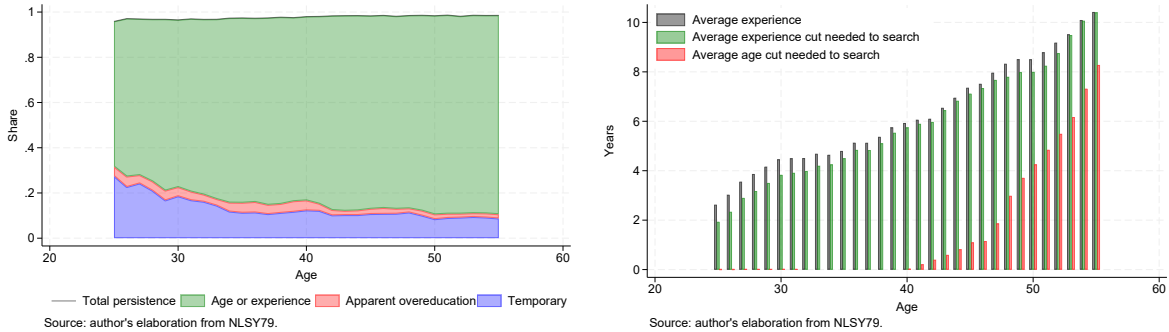
the case whenever the calibrated model predicts a positive matching probability⁷, given the worker’s type. The “apparent overeducation” channel is assigned to humanities graduates who are overeducated in jobs with complexity r_3 since, given their education type, they are in the best possible job (they do not satisfy the minimum requirement for r_4). If none of these two channels apply, the persistence is due to specialization and/or age. In this case, I will first assign a generic “age/experience” channel to the overeducated worker. Then, I will disentangle whether the persistence is mainly due to age or specialization. In particular, the minimum combination of age and experience reduction that would make the worker search for another occupation is computed. For example, if a 40-year-old worker with 10 years of experience would search for another occupation if she were 38 years old with 8 years of experience, then the age/experience reduction is 2 years of age and 2 years of experience.

⁷Actually, when the optimal θ is higher than a very low threshold (0.000001), which corresponds to a matching probability of 0.1%.

4.4 Decomposition of Overeducation Persistence: Results

In this section, I present and discuss the results of decomposing the persistence of overeducation displayed in Figure 9. First, the persistence is partially temporary, and this channel becomes less relevant as workers get older, as expected. In particular, at the beginning of the working lifecycle, approximately 25 percentage points of the total monthly persistence are due to search frictions. This channel slowly decreases to around 10 percentage points and remains stable afterwards. The “apparent overeducation” channel plays a minor role, accounting for around 5 percentage points of the total persistence. This magnitude does not change significantly throughout the lifecycle.

Figure 9: Decomposition of monthly overeducation persistence (NLSY79).



Note: This figure reports the decomposition of monthly overeducation persistence into the following channels: (i) **temporary** (search frictions), (ii) **apparent overeducation**, (iii) **specialization** (job-specific experience), and (iv) **age** (skill depreciation and a shorter time horizon). The left panel shows the lifecycle evolution of each channel’s contribution, with age and specialization grouped together; the right panel isolates the joint roles of age and experience by showing the smallest reduction in age and experience needed to make it optimal for the worker to search for another occupation. Results are computed from the model calibrated to the NLSY79. See Section 4 for details on the calibration strategy.

The rest of the persistence in the first half of the lifecycle is almost entirely due to the job-specific experience, which leads overeducated workers to specialize in the current occupation. This means that overeducated workers accumulate experience that they do not want to lose by moving to a potentially more productive occupation. This mechanism is amplified by ability learning frictions, which add more uncertainty, the slower the learning process. Experience remains the major channel also in the second half of the lifecycle, but now age plays a more important role. In particular, after age 40, a mix between age and experience is the main driver of persistence.

One may argue that the assumption of complete non-transferability drives the results. In Appendix A (Figure A1), I relax this assumption by allowing workers to retain 50% of their specific experience upon switching occupations. I find that while search frictions become the dominant driver of overeducation for young workers, the specialization trap remains the key mechanism for persistence in the long run.

Overall, the decomposition highlights a clear lifecycle pattern. Search frictions are mainly relevant at the beginning of careers, while specialization, reinforced later by age, drives long-run persistence.

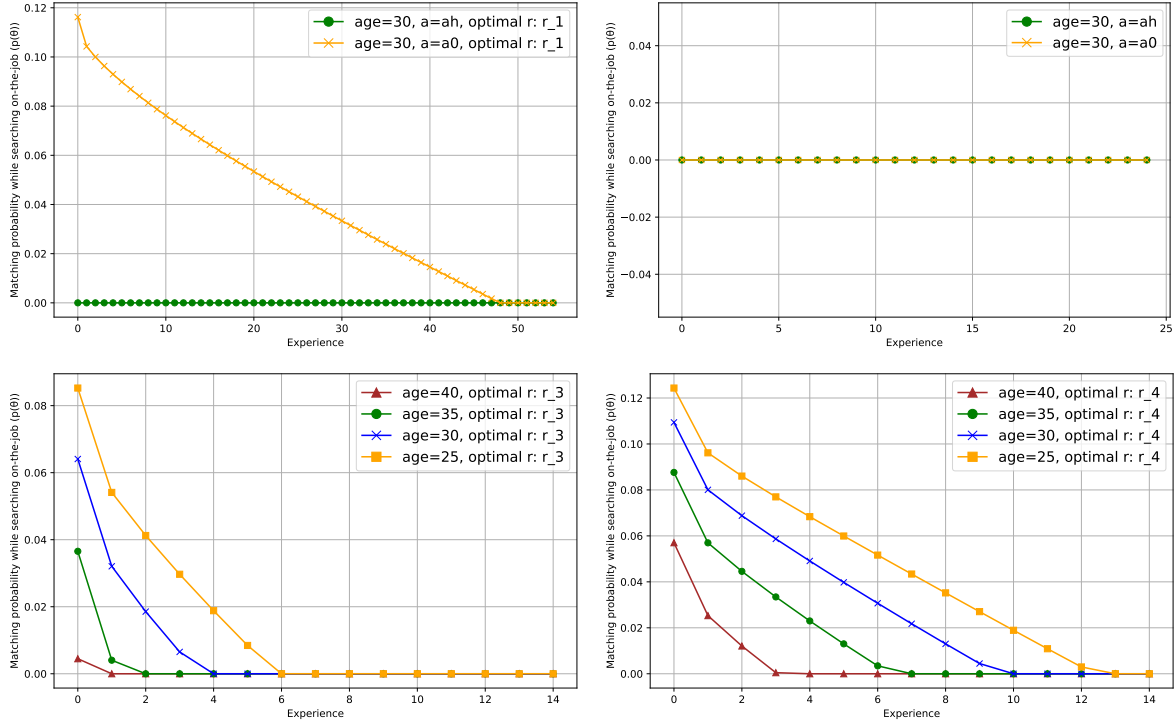
4.5 Learning Frictions for Low-educated Workers

Learning frictions can be potentially highly impactful on the workers' optimal choice. To see this, the top-left panel of Figure 10 illustrates the optimal choice of low-educated workers (E^L) employed in jobs with complexity r_2 . When ability is revealed to be high ($a_{ij} = a^H$), workers cease searching for alternative occupations: despite the possibility of moving to a more complex job (r_3), the risk associated with learning frictions makes it optimal to remain in the current match. Very different is the situation when the ability is unknown ($a_{ij} = a^0$). In this case, workers do search for an alternative occupation and the learning frictions are so heavy for them that they prefer to search for a new occupation with a lower complexity, r_1 , where they are sure to satisfy the skill requirement. In other words, the fact that their innate ability in the current occupation is unknown (and there is a high probability, around 60%, that $a_{ij} = a^L$) makes it optimal to search for jobs in the lowest complexity segment of the labor market and specialize in one of them. As they get more experienced, conditional on not having found another job due to search frictions, they have a higher opportunity cost of leaving their current occupation and, so, they search in submarkets where the employment contract value, x , is higher and the probability of matching, $p(\theta)$, is lower.

4.6 Learning Frictions for College Graduates

One may be concerned that a similar process also holds for college graduates employed in jobs with complexity r_3 when ability is unknown. However, Figure 10 (top-right panel) shows that this is not the case. While employed in these kinds of jobs, college graduates do not search for alternative occupations, but they prefer to wait until they discover their ability level in the current occupation. The main difference with non-college graduates lies in the belief of being high-ability; this is much higher both for the humanities graduates (66.9%) and for non-humanities graduates (73.4%) compared to non-college graduates (41.1%). Therefore, the expected value of waiting and having the possibility of staying employed in a more complex occupation is high enough to outweigh the potential loss given by the possibility of being low-ability in the current occupation and restarting from zero in another occupation. Notice that in theory, non-humanities graduates could search for a more complex occupation in r_4 , but the potential gain is dominated by the job-specific experience loss, so that they prefer to stay in the current occupation.

Figure 10: Matching probability while searching on the job for different worker types and job complexities.

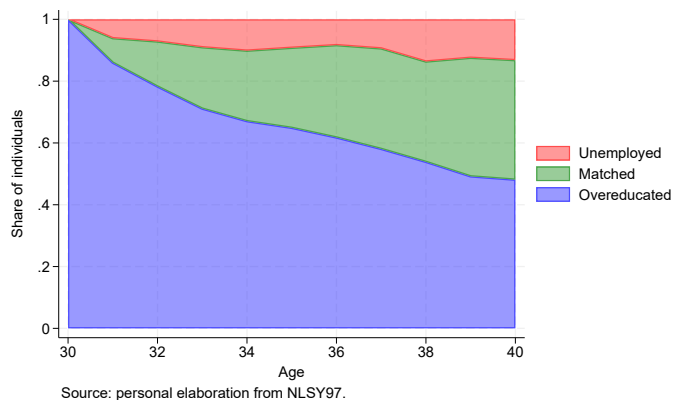


Note: Each panel reports the optimal on-the-job matching probability $p(\theta)$ on the y-axis and occupation-specific experience on the x-axis for a specific worker type and job complexity. The optimal r reported in the legend indicates the complexity level of the occupation where the worker is searching for a job while employed. Top-left: workers with no college degree (E^L) employed in jobs of complexity r_2 , aged 30, known high innate ability a^H vs unknown ability a^0 . Top-right: workers with a college degree (applies to both humanities and non-humanities) employed in jobs of complexity r_3 , aged 30, known high innate ability a^H vs unknown ability a^0 . Bottom-left: college graduates in humanities employed in jobs of complexity r_2 , different ages. Bottom-right: college graduates in non-humanities employed in jobs of complexity r_2 , different ages.

4.7 College Graduates Optimal Choice in Lower Complexity Jobs

In this section, I discuss the optimal behavior of college graduates in relatively low complexity jobs r_2 , the complexity level with the most overeducated workers. Figure 10 (bottom panels) shows their optimal matching probability, $p(\theta)$, when searching on-the-job. Both experience and age play an important role in determining the optimal choice of workers. In particular, for 30-year-old humanities graduates, it is optimal to search for an alternative occupation in complexity r_3 only within the first 6 months of employment in the current occupation. After this threshold, the "specialization" effect keeps workers attached to their current job (probably overeducated). The same pattern is even stronger for older workers, who have a shorter time horizon to enjoy

Figure 11: Evolution of the status of overeducated workers at age 30 (NLSY97).



Note: This figure is the analogue of Figure 1 for the younger cohort and follows all the individuals in the sample who are classified as *overeducated* at age 30 (NLSY97). The x-axis reports the age starting from 30; the y-axis shows the fraction of the original subsample (normalized to 100%) in each subsequent status. Series shown are: (i) **remaining overeducated** (blue area), that is, the share still employed in occupations whose typical educational requirement (from O*NET) is lower than the worker's attainment; (ii) **transitioned to matched** (green area), that is, the share employed in occupations whose typical requirement matches the worker's education; (iii) **transitioned to unemployment** (red area), that is, the share currently unemployed. Overeducation is defined as having attained a higher education level than the occupation's typical requirement. See Section 2.1 for sample restrictions and further details. See Section 2.3 for details on the overeducation definition.

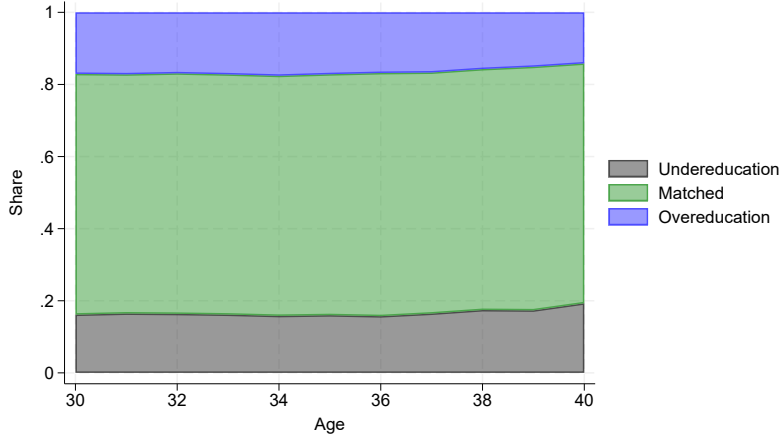
the risky choice of moving to another occupation (if $a_{ij} = a^H$ in the new occupation) or to mitigate the loss of an eventually bad choice (if $a_{ij} = a^L$ in the new occupation). The same argument applies to non-humanities college graduates, but they search for a longer period, up to 12 months for 30-year-old workers, and with more intensity. This is the case because they have a higher probability of being high-ability in the new occupation, and they can search for more complex jobs (r_4) in which they are slightly more productive.

5 Comparison with Different Cohort (NLSY97)

To assess the robustness of the results, I compare the baseline findings from the NLSY79 (older cohort) with those obtained using the NLSY97 (younger cohort). More specifically, the NLSY97 cohort refers to individuals born between 1980 and 1984.

Given that this newer cohort is very recent, individuals are around 40 years old at most, so only results about the first half of the lifecycle can be discussed. However, this is probably the most relevant part of the analysis, as it is more challenging to break down the potential inefficiencies due to overeducation as workers age. First, Figure 11 is the analogue of Figure 1 for the younger cohort. The pattern appears similar, although the persistence is slightly more pronounced. Indeed, after 5 years, more than

Figure 12: Share of workers in each status by age (NLSY97).



Source: personal elaboration from NLSY97.

Note: This figure is the analogue of Figure 3 (right panel) for the younger cohort (NLSY97) and shows the share of workers in each status by age. See Section 2.1 for sample restrictions and further details. See Section 2.3 for details on the overeducation/matched/undereducation definitions.

60% of overeducated workers at 30 are still experiencing the same condition (it was around 50% in NLSY79). After 10 years, almost 50% are still classified as overeducated (40% in NLSY79).

Second, the overeducation status in Figure 12 is very similar as well, showing both the share of overeducated and undereducated around 18% at 30 years old, followed by a little decrease (increase) in the share of overeducated (undereducated) before reaching 40 years old.

Finally, Figure 13 shows the decomposition of overeducation persistence. The key result is the same as in the baseline dataset: in the first half of the career, specialization is the most important driver leading to overeducation persistence. Here, the temporary channel is even less relevant, suggesting that overeducation is becoming more persistent. Moreover, age and apparent overeducation also have a minor role in this case, at least in the first half of the lifecycle.

Overall, the comparison confirms that the key mechanisms identified in the model are robust across cohorts.

6 Conclusion

This paper has examined the persistence of overeducation over the lifecycle through the lens of a directed search model with heterogeneous workers and occupations. Using U.S. data from the NLSY79 and NLSY97, the model successfully replicates the empirical patterns of overeducation persistence and provides a structural decomposition of its

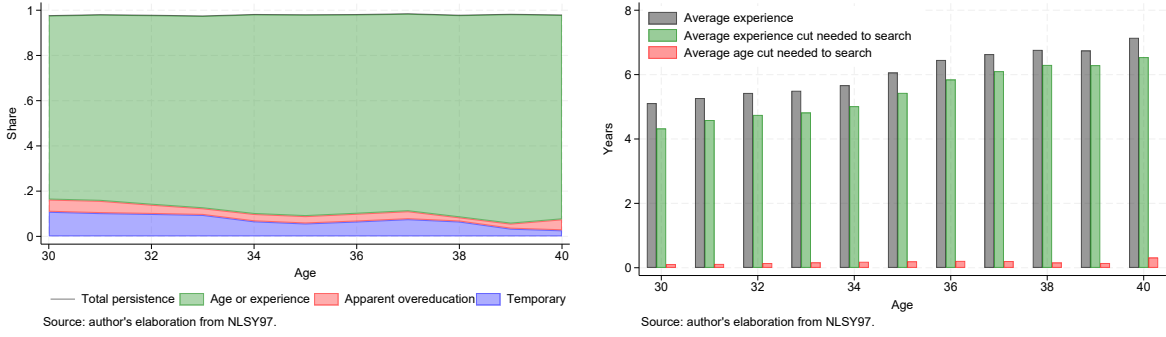


Figure 13: Decomposition of monthly overeducation persistence (NLSY97).

Note: This figure is the analogue of Figure 9 for the younger cohort (NLSY97) and reports the decomposition of monthly overeducation persistence into the following channels: (i) **temporary** (search frictions), (ii) **apparent overeducation**, (iii) **specialization** (job-specific experience), and (iv) **age** (skill depreciation and a shorter time horizon). The left panel shows the lifecycle evolution of each channel’s contribution, with age and specialization grouped together; the right panel isolates the joint roles of age and experience by showing the smallest reduction in age and experience needed to make it optimal for the worker to search for another occupation. Results are computed from the model calibrated to the NLSY79 and applied to the NLSY97 cohort. See Section 4 for details on the calibration strategy.

underlying mechanisms.

Three main insights emerge. First, labor market frictions explain why many workers remain temporarily overeducated early in their careers, but their role fades with age. Second, specialization, driven by the accumulation of occupation-specific experience, emerges as the dominant force behind long-run persistence. Third, the interaction between specialization and age further amplifies this persistence in later stages of the lifecycle.

These findings highlight that overeducation is not merely a transient phenomenon but can result from structural mechanisms that lock workers into mismatched jobs. Policy implications follow directly from the mechanisms identified. First, because search and especially learning frictions make early mismatches persistent as workers accumulate non-transferable job experience, policies that accelerate learning and improve early-career matching can materially reduce long-run overeducation. Second, incentives that lower the individual cost of moving to more complex occupations help undo specialization-driven traps. Finally, one may consider reducing the extent of the mismatch in the first place when entering the labor market, for example through policies that aim to reduce search frictions and pushing individuals towards jobs more aligned with their education right after graduation. Overall, a lifecycle perspective is essential for understanding and addressing the persistence of overeducation.

References

- Alba-Ramirez, A. (1993). Mismatch in the spanish labor market: overeducation? *Journal of Human Resources*, pages 259–278.
- Baley, I., Figueiredo, A., and Ulbricht, R. (2022). Mismatch cycles. *Journal of Political Economy*, 130(11):2943–2984.
- Bauer, T. K. (2002). Educational mismatch and wages: a panel analysis. *Economics of education review*, 21(3):221–229.
- Becker, G. S. (1964). Human capital: A theoretical and empirical analysis with special reference to education. In *Human Capital: A Theoretical and Empirical Analysis with Special Reference to Education, First Edition*. The University of Chicago Press.
- Cappelli, P. H. (2015). Skill gaps, skill shortages, and skill mismatches: Evidence and arguments for the united states. *ILR review*, 68(2):251–290.
- Chevalier, A. (2003). Measuring over-education. *Economica*, 70(279):509–531.
- Clark, B., Joubert, C., and Maurel, A. (2017). The career prospects of overeducated americans. *IZA Journal of Labor Economics*, 6:1–29.
- Dolado, J. J., Jansen, M., and Jimeno, J. F. (2009). On-the-job search in a matching model with heterogeneous jobs and workers. *The Economic Journal*, 119(534):200–228.
- Dolton, P. and Vignoles, A. (2000). The incidence and effects of overeducation in the uk graduate labour market. *Economics of education review*, 19(2):179–198.
- Duncan, G. J. and Hoffman, S. D. (1981). The incidence and wage effects of overeducation. *Economics of education review*, 1(1):75–86.
- Freeman, R. B. (1975). Overinvestment in college training? *Journal of human resources*, pages 287–311.
- Gautier, P. A. (2002). Unemployment and search externalities in a model with heterogeneous jobs and workers. *Economica*, 69(273):21–40.
- Gautier, P. A., Van den Berg, G. J., Van Ours, J. C., and Ridder, G. (2002). Worker turnover at the firm level and crowding out of lower educated workers. *European Economic Review*, 46(3):523–538.
- Guvenen, F., Kuruscu, B., Tanaka, S., and Wiczer, D. (2020). Multidimensional skill mismatch. *American Economic Journal: Macroeconomics*, 12(1):210–244.

- Hall, R. E. and Milgrom, P. R. (2008). The limited influence of unemployment on the wage bargain. *American economic review*, 98(4):1653–1674.
- Hartog, J. (2000). Over-education and earnings: where are we, where should we go? *Economics of education review*, 19(2):131–147.
- Ichino, A., Rustichini, A., and Zanella, G. (2024). College, cognitive ability, and socioeconomic disadvantage: policy lessons from the uk in 1960-2004.
- Korpi, T. and Tåhlin, M. (2009). Educational mismatch, wages, and wage growth: Overeducation in sweden, 1974–2000. *Labour economics*, 16(2):183–193.
- Leuven, E. and Oosterbeek, H. (2011). Overeducation and mismatch in the labor market. *Handbook of the Economics of Education*, 4:283–326.
- Lise, J. and Postel-Vinay, F. (2020). Multidimensional skills, sorting, and human capital accumulation. *American Economic Review*, 110(8):2328–2376.
- McGuinness, S., Pouliakas, K., and Redmond, P. (2018). Skills mismatch: Concepts, measurement and policy approaches. *Journal of Economic Surveys*, 32(4):985–1015.
- Menzio, G. and Shi, S. (2011). Efficient search on the job and the business cycle. *Journal of Political Economy*, 119(3):468–510.
- Menzio, G., Telyukova, I. A., and Visschers, L. (2016). Directed search over the life cycle. *Review of Economic Dynamics*, 19:38–62.
- Moen, E. R. (1997). Competitive search equilibrium. *Journal of political Economy*, 105(2):385–411.
- Montgomery, J. D. (1991). Equilibrium wage dispersion and interindustry wage differentials. *The Quarterly Journal of Economics*, 106(1):163–179.
- Roth, B., Becker, N., Romeyke, S., Schäfer, S., Domnick, F., and Spinath, F. M. (2015). Intelligence and school grades: A meta-analysis. *Intelligence*, 53:118–137.
- Rubb, S. (2003). Overeducation: a short or long run phenomenon for individuals? *Economics of education review*, 22(4):389–394.
- Sicherman, N. (1991). ” overeducation” in the labor market. *Journal of labor Economics*, 9(2):101–122.
- Sicherman, N. and Galor, O. (1990). A theory of career mobility. *Journal of political economy*, 98(1):169–192.

Spence, M. (1978). Job market signaling. In *Uncertainty in economics*, pages 281–306. Elsevier.

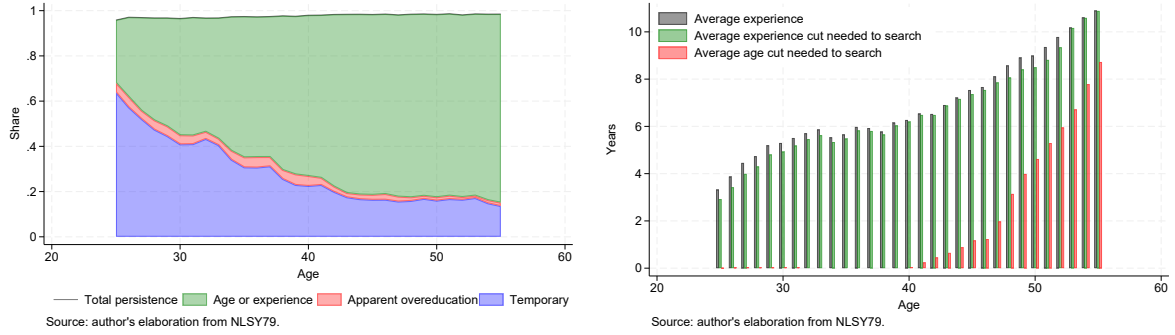
Strenze, T. (2007). Intelligence and socioeconomic success: A meta-analytic review of longitudinal research. *Intelligence*, 35(5):401–426.

Verdugo, R. R. and Verdugo, N. T. (1989). The impact of surplus schooling on earnings: Some additional findings. *Journal of human resources*, pages 629–643.

Appendix

A Additional Figures

Figure A1: Decomposition of monthly overeducation persistence (NLSY79, experience partially retained).



Note: This decomposition is the result of an alternative specification where workers retain half of their job-specific experience when switching occupations. This figure reports the decomposition of monthly overeducation persistence into the following channels: (i) **temporary** (search frictions), (ii) **apparent overeducation**, (iii) **specialization** (job-specific experience), and (iv) **age** (skill depreciation and a shorter time horizon). The left panel shows the lifecycle evolution of each channel's contribution, with age and specialization grouped together; the right panel isolates the joint roles of age and experience by showing the smallest reduction in age and experience needed to make it optimal for the worker to search for another occupation. Results are computed from the model calibrated to the NLSY79. See Section 4 for details on the calibration strategy.

B Proof of Block Recursive Equilibrium

The steps are similar to the proof of Theorem 1 in Menzio et al. (2016).

Let's proceed by backward induction. In the last period T , the value of unemployment is

$$U_T(\omega, \psi) = b$$

The joint value of the match is

$$V_T(\omega, \psi) = \begin{cases} y(r, e, s, t) & \text{if } a \neq a^0 \\ \mathbb{E}[y(r, e, s, t)|E] & \text{if } a = a^0 \end{cases}$$

Notice that both U_T and V_T do not depend on the state ψ , hence $U_T(\omega, \psi) = U_T(\omega)$ and $V_T(\omega, \psi) = V_T(\omega)$.

The market tightness must satisfy

$$\theta_T(\omega, x, \psi) = \begin{cases} q^{-1}\left(\frac{k}{V_T(\omega) - x}\right) & \text{if } x \leq V_T(\omega) - k \\ 0 & \text{otherwise} \end{cases}$$

Notice that $\theta_T(\omega, x, \psi) = \theta_T(\omega, x)$.

The problem of the unemployed is

$$R_T(\omega, \psi) = \max \left\{ \max_x \{p(\theta_T(\omega, x))[x - U_T(\omega)]\}, \max_{x,r} \{p(\theta_T(\omega_0, x))[x - U_{t+1}(\omega)]\} \right\}$$

Let's start from the second term, notice that $x = V_T(\omega) - \frac{k}{q(\theta_T(\omega, x))}$ whenever $x \leq V_T(\omega) - k$ and $\theta_T(\omega, x) = 0$ otherwise. So the second term can be rewritten as

$$\begin{aligned} \max_{x,r} \{p(\theta_T(\omega_0, x'))[V_T(\omega_0) - \frac{k}{q(\theta_T(\omega_0, x))} - U_T(\omega)]\} = \\ = \max_{x,r} \{-k\theta_T(\omega_0, x) + p(\theta_T(\omega_0, x))[V_T(\omega_0) - U_T(\omega)]\} \end{aligned}$$

Now, fixing r we can compute for each $\bar{r} \in \{r_1, r_2, r_3, r_4\}$

$$\begin{aligned} \max_x \{-k\theta_T(\omega_0(\bar{r}), x) + p(\theta_T(\omega_0(\bar{r}), x))[V_T(\omega_0(\bar{r})) - U_T(\omega)]\} = \\ = \max_{\theta} \{-k\theta + p(\theta)[V_T(\omega_0(\bar{r})) - U_T(\omega)]\} \end{aligned}$$

The maximizer of this problem gives the optimal choice for complexity $\bar{r}$, let's denote this maximizer as $\theta_{\bar{r}}^*$. Once we have computed this for each complexity level, we can compare the value function for each of the 4 maximizers to obtain the optimal choice for the second term (optimal new occupation), let's denote it as θ_{new}^* . Similarly, the first term can be rewritten as

$$\begin{aligned} \max_x \{p(\theta_T(\omega, x))[V_T(\omega) - \frac{k}{q(\theta_T(\omega, x))} - U_T(\omega)]\} = \\ = \max_x \{-k\theta_T(\omega, x) + p(\theta_T(\omega, x))[V_T(\omega) - U_T(\omega)]\} = \\ = \max_{\theta} \{-k\theta + p(\theta)[V_T(\omega) - U_T(\omega)]\} \end{aligned}$$

Let's denote the solution of this problem as θ_{old}^* . The policy function of the unemployed is $\theta_T^u = \theta_{old}^*$ if the optimal first term is better than the optimal second term and $\theta_T^u = \theta_{new}^*$ otherwise. Notice that $\theta_T^u(\omega, \psi) = \theta_T^u(\omega)$ and $R_T(\omega, \psi) = R_T(\omega)$. Also, we denote as $r_T^u(\omega), e_T^u(\omega), a_T^u(\omega)$ the policy functions for the complexity, experience and ability associated with $\theta_T^u(\omega)$. It follows

$$x_T^u(\omega) = V_T(E, r_T^u, exp_T^u, a_T^u) - \frac{k}{q(\theta_T^u(\omega))}$$

A similar reasoning holds for the search problem of the employed

$$S_T(\omega, \psi) = \max_{x, r} \{p(\theta_T(\omega_0, x, \psi)) [x - \mathbb{E}_{\hat{a}|a} V_T(\omega)]\}$$

Following the same steps for the second term of the unemployed problem above, it is possible to compute $\theta_T^e(\omega, \psi) = \theta_T^e(\omega)$, $S_T(\omega, \psi) = S_T(\omega)$, $r_T^e(\omega)$ and

$$x_T^e(\omega) = V_T(E, 0, r_T^e, a^0) - \frac{k}{q(\theta_T^e(\omega))}$$

The employment policy function d_T must solve

$$\max_{d \in [\delta, 1]} \{dU_T(\omega) + (1 - d)[V_T(\omega) + \lambda_e S_T(\omega)]\}$$

Clearly $d_T = 1$ if $U_T(\omega) > [V_T(\omega) + \lambda_e S_T(\omega)]$ and $d_T = \delta$ otherwise. Also, $d_T(\omega, \psi) = d_T(\omega)$.

We can move to period $T - 1$, where we have

$$\begin{aligned} U_{T-1}(\omega, \psi) &= b + \beta[U_T(\hat{\omega}) + \lambda_u R_T(\hat{\omega})] \\ V_{T-1}(\omega, \psi) &= y(r, e, s, t) + \beta\{d_T(\omega)U_T(\hat{\omega}) + (1 - d_T(\omega))[\mathbb{E}_{\hat{a}|a} V_T(\hat{\omega}) + \lambda_e S_T(\hat{\omega})]\} \\ \theta_{T-1}(\omega, x, \psi) &= \begin{cases} q^{-1}\left(\frac{k}{V_{T-1}(\omega) - x}\right) & \text{if } x \leq V_{T-1}(\omega) - k \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Also in this case, the value functions do not depend on the state ψ . By repeating the same steps as above for all periods, we get the unique BRE.

C Measurement Details: Complexity and Ability Construction

In this appendix, I provide a detailed description of the procedure used to construct the job complexity index (r_j) and the binary measure of worker’s innate ability (a_{ij}), complementing the summary provided in Section 2.2.

C.1 Job Complexity Construction

The starting point is the set of four percentile scores for each occupation j derived from the Principal Component Analysis (PCA) on O*NET descriptors performed in Baley et al. (2022): verbal (v_j), mathematical (m_j), technical (t_j), and social (s_j). Let $\mathcal{D}_j = \{v_j, m_j, t_j, s_j\}$ be the set of these scores for occupation j , where each element represents the percentile rank of the occupation in the distribution of skill requirements.

To capture the complexity, I compute the *effective complexity score*, C_j , as the average of the three highest dimensions:

$$C_j = \frac{1}{3} \sum_{k=1}^3 d_{(k),j} \quad (2)$$

where $d_{(k),j}$ denotes the k -th highest value in the set $\mathcal{D}_j$. The reason why I focus on the top three dimensions is to account for the fact that occupations may require a combination of skills, and reflects the idea that job complexity is not solely determined by a single skill but rather by a combination of multiple skills. At the same time, this approach does not overly penalize occupations that may have a lower requirement in one dimension but are still complex due to high demands in the other dimensions. Based on the distribution of C_j , occupations are categorized into four discrete complexity levels $r_j \in \{r_1, r_2, r_3, r_4\}$ using the following thresholds:

$$r_j = \begin{cases} r_1 & \text{if } C_j \leq 15 \\ r_2 & \text{if } 15 < C_j < 50 \\ r_3 & \text{if } 50 \leq C_j < 85 \\ r_4 & \text{if } C_j \geq 85 \end{cases} \quad (3)$$

Finally, to ensure consistency with the theoretical framework where the highest complexity level (r_4) is accessible only to college graduates, I apply a correction: any occupation with $C_j \geq 85$ that does not typically require a college degree (based on O*NET education requirements) is reclassified as r_3 .

C.2 Worker’s Innate Ability Construction

For each worker i and occupation j , innate ability a_{ij} is a binary variable taking values $\{a^L, a^H\}$. This is determined by comparing the worker’s skill vector, derived from ASVAB and psychological scores, with the occupation’s requirement vector.

Let S_{ik} denote worker i ’s percentile score in skill dimension $k \in \{\text{math, verbal, technical, social}\}$, and let R_{jk} denote the requirement percentile of occupation j in the same dimension. A worker is considered to have high innate ability ($a_{ij} = a^H$) in occupation j if their skills satisfy the requirements in *all* four dimensions, subject to a tolerance parameter τ :

$$a_{ij} = \begin{cases} 1 \ (a^H) & \text{if } S_{ik} \geq \tau \cdot R_{jk} \quad \forall k \in \{M, V, T, S\} \\ 0 \ (a^L) & \text{otherwise} \end{cases} \quad (4)$$

In the baseline calibration, the tolerance parameter is set to $\tau = 0.5$. This implies that a worker is suitable for the job (high ability) if they possess at least 50% of the required skill level in every dimension. The strict AND condition ($\forall k$) captures the idea that a significant deficiency in any single key dimension can hinder productivity, regardless of proficiency in others. However, given the tolerance parameter, workers can still be classified as high-ability even if they do not fully meet the requirements in all dimensions, as long as they meet at least half of the requirements.